SULIT

TAKE HOME EXAMINATION: MIDTERM EXAMINATION
ESSION 2020/2021, SEMESTER 1

SUBJECT CODE	**: SCSJ 3323**
SUBJECT NAME	**: SOFTWARE DESIGN AND ARCHITECTURE**
YEAR/COURSE	**: SCSJ**
TIME	**: 5 HOURS**
DATE	**: 5th DECEMBER 2020, 9.00 AM – 2.00 PM**
VENUE	**: NOT APPLICABLE**

SECTION A: Introduction to Software Engineering Design & Software Design with UML (Chapter 1 and 2) (40 Marks)

SECTION B: Principles of Software Architecture, Patterns and Style in Software Architecture & Principles of Detailed Design (Chapter 3, 4 and 5) (60 Marks)

TOTAL: 100 MARKS

INSTRUCTION AND REMINDER

Answer all questions in the given template. All diagrams must be drawn using the UML diagram drawing tools. Note that any form of plagiarism will cause students to get zero in this assessment. Submit the assignment via e-Learning or as instructed by the course lecturer. This is the Take Home Examination to replace the Midterm exam that is not conducted due to the COVID-19 pandemic and Movement Control Order (MCO) since 18 March 2020. This examination contributes 20% of the whole course assessments.

Name	
IC No. / Matric No.	
Year / Programme	
Section	
Lecturer Name	

This exam consists of EIGHT (8) pages excluding this page.

SECTION A - STRUCTURE QUESTIONS

INSTRUCTIONS: Answer all of the following TWO (2) questions.

Question 1 **(20 marks)**

a) Software design is an important process in software development lifecycle. However, a lot of software developer/house tends to ignore the process. Therefore, provide and explain 2 motivations on software design.

(4 marks)

b) Two major activities of the software design process include Software Architecture and Detailed Design. Demonstrate on how these two activities being executed?

(6 marks)

Case study on Online Counselling system

Due to the current COVID-19 pandemic in Malaysia, only essential sectors such as pharmacies and banking can operate. Other sectors need to find suitable alternatives in order to continue their services. Assume you are attached to a software house (company) that is responsible to solve the issues related to traditional face-to-face activities that now need to be done online.

As the corona virus pandemic rapidly sweeps across the world, it is inducing a considerable degree of fear, worry and concern in the population at large and among certain groups in particular, such as older adults, care providers and people with underlying health conditions. In public mental health terms, the main psychological impact to date is elevated rates of stress or anxiety. But as new measures and impacts are introduced – especially quarantine and its effects on many people's usual activities, routines or livelihoods – levels of loneliness, depression, harmful alcohol and drug use, and self-harm or suicidal behaviour are also expected to rise.

In populations already heavily affected, issues of service access and continuity for people with developing or existing mental health conditions are also now a major concern, along with the mental

health and well-being of front-line workers. As part of its public health response, you and your software house will construct an online web counselling system (e-Counselling) in order to assist on the mental health and psycho-social support aspects of COVID-19 pandemic.

The users of e-Counselling include patients, administrator and counsellor. The administrator can register new counsellors and delete registered counsellors on the system while patients can register themselves but need administrator to approve the patients' accounts. In order to get a clear overview of the patients background, the counsellor can have access and analyse on social data stored from any social media account that the patients have such as Facebook, Instagram or Twitter. Therefore, the patients can register to e-Counselling system using their media social accounts. Then, the administrator can assign counsellor to patient by default or by patients' requests. Lastly the administrator can have the log report of mail notification sending from both patients and counsellor. The counsellors and patient can update their profile data that have been registered by the administrator.

Confidentiality of counseling content is important and the content is strictly secret, thus the system must prevent information from leaking. Counseling took place by both patients and counselors sending message using e-Counselling system. All messages are stored in a database of the system to prevent the information from leaking outside. Then, the system automatically sends notification email to counselors and patients. After that, read and respond message for consultation after login to e-Counselling. The reason why email is just used for notification is to prevent the counseling content leaked by intercepting mail server communication or computer virus. Last but not least, the patient and the assigned counselor can have online chat with each other at any time both of them are online.

c) Name **FIVE**(5) design principles that should be considered during developing the system on the provided case study.

(10 marks)

Question 2 **(20 marks)**

a) Draw a class diagram based on Table 1 below:

Table 1: List of Classes, Operations, Attributes and Relationships

Class Name		**Operations**			**Attributes**			**Relationships**
Name	**Type**	**Return**	**Name**	**Visibility**	**Return**	**Name**	**Visibility**	
ClassPayment	Interface	Void	Type	Public	int	TypePayment	private	Associate with ClassBooking using a directional association
		Double	PaymentAmount	Public	Double	Amount	Protected	
ClassBank	Class	Void	Type	Public	int	TypePayment	private	Inherits from ClassPayment
		Double	PaymentAmount	Public	Double	Amount	Protected	
ClassBooking	Class	void	createbooking	public	int	BookingID	Private	Associate with ClassPayment using a directional association
		bool	confirmPayment	public	Int	ItemID	Protected	
					bool	confirm	Protected	
ClassCOD	Class	Void	Type	Public	int	TypePayment	private	Inherits from ClassPayment
		Double	PaymentAmount	Public	Double	Amount	Protected	
ClassCustome	Class	void	makePayment	Public	int	CustomerID	Private	Aggregation with ClassBooking
		void	chooseItem	Public	String	Name	Protected	

(10 marks)

b) Draw the following UML diagrams based on the provided case study

i. Use Case Diagram. Must provide the appropriate <<include>> and/or <<extension>> relationships

(6 marks)

ii. Sequence Diagram. Choose to draw a sequence diagram from the use case that could include at least **ONE (1)** boundary and **TWO (2)** entities. Show the messages clearly.

(4 marks)

SECTION B - ESSAY / PROBLEM SOLVING

INSTRUCTIONS: Answer ALL of the following questions based on the provided SRS template.

Question 3 **(10 marks)**

a) You are the project manager of a team to develop the project based on the given SRS of the project. As a project manager how would you suggest to ensure a faithful transformation of the project from requirements to detailed design using what you have learned in the Software Design and Architecture course **(5 marks)**

b) Your project manager given you the SRS of the project. You are given the task to identify the core requirements specification from the project. You have to apply the concept of complete, consistent, attainable requirements statement in order to come out with FOUR (4) important functional requirements statements and ONE (1) non-functional requirement statement.

(5 marks)

Question 4 **(12 marks)**

a) Using the same SRS as a reference, apply the concept of physical view and process view and explain how would you use the information in the SRS in order to construct both views.

(4 marks)

b) Create process view based on the information identified in the given SRS.

(4 marks)

c) Create physical view based on the information identified in the given SRS.

(4 marks)

Question 5 **(13 marks)**

a) As you are working on the SRS, you have identified several quality attributes for the project to be developed. From the initial discussion, the development team are in a hot discussion of whether to choose pipe or filter architecture pattern or layered architecture pattern for the project. Your project manager has appointed you to come out with the strongest justification for the chosen architecture pattern for the project.

Give **FOUR (4)** strongest argument or justification of why that particular architecture pattern should be chosen.

(8 marks)

b) Create a logical view of the selected architecture pattern that you identified earlier in (a)

(5 marks)

Question 6 **(25 marks)**

As a software engineer in her team, Asiah was given a task to model the different types of utilities in the UTILITY management system. Originally, the design has only two types of utilities, i.e. Space and Equipment. Space class has two operations i.e. doBooking() and payRent(). Equipment class has one operation onLoan(). However, there is a need to add a new utility, SportCenter that have both operations from Space and Equipment class.

Therefore, Asiah has proposed a new design that adds a new utility which is SportCenter which can perform all operations, as shown in Figure 6.1. As a consequence, Space class and Equipment class inherit operations that they never need (and never use). The design introduces a bloated interface problem.

a) By applying the Interface Segregation Principle (ISP), redraw the diagram to solve Asiah's bloated interface problem.

(5 marks)

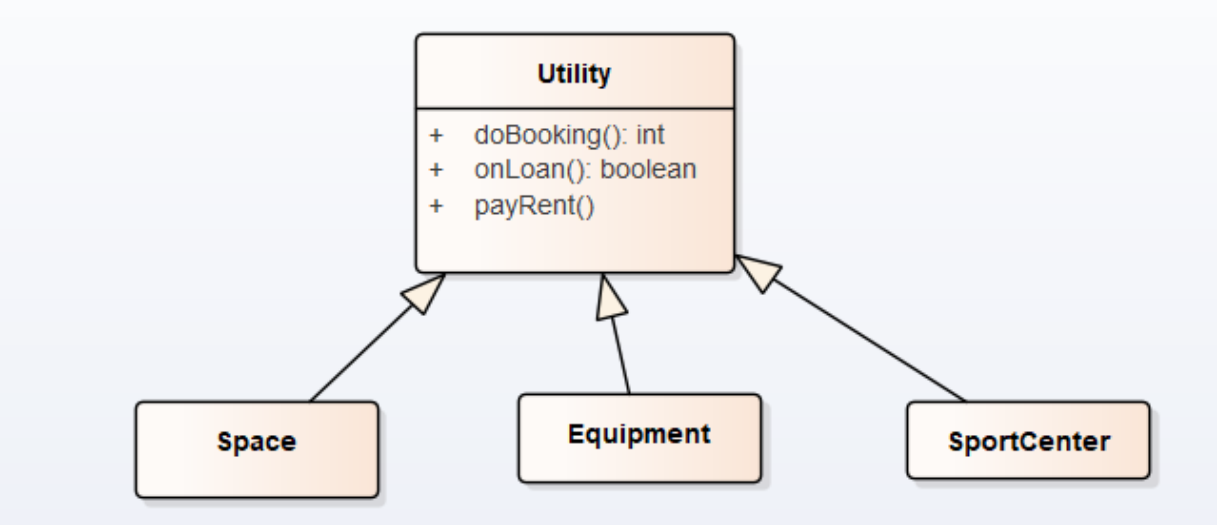


Figure 6.1 Utility

b) This question is based on Manage Utility use case which can be referred in section **2.2.13** from the SRS of UTILITY Management System and DO NOT relate with the solution you have done in 6 (a). The additional requirements are described below:

There are 3 types of Utility managed by the Admin of UTILITY management system namely Space, Equipment and GymCenter.

Each Utility type has 3 common operations, doBooking(), payRent() and cancelBooking(). However, each utility has different behaviours depending on its type

i. **doBooking()** operation: to book for Space the booking must be for 1 day or more while for Equipment the booking must be for 2 days or more . However, for the GymCenter the booking must be based on hours.
ii. **payRent()** operation: to pay rent for Space its must be based on the days of booking while for Equipment the rent must be paid based on the down payment and also days of booking. Rate per day is RM10. As for GymCenter the rent must be paid based on how many hours of the booking and also the down payment. Rate per hour is RM5.
iii. **cancelBooking**() operation: to cancel booking for Space and Equipment the booking can be cancelled 2 days before else the booking could not be cancelled. As for the GymCenter, the booking cancellation should be done 5 hours before the actual time or else the booking could not be cancelled.

Currently, there are only three types of Utility, as described in the above requirements. However, it is expected that different types of utilities will be added in the future. To solve this problem, you are required to apply the Open-Closed Principle (OCP) in your detailed design.

Draw a class diagram that applies OCP to solve the above problem.

(10 marks)

c. Write the implementation (either using C#, Java or C++ language) of the doBooking(), payRent(), and cancelBooking() operations for each class that represents the Utility types as described in Question 6(b). Identify what are the suitable attributes for the classes which are Space, Equipment and GymCenter and suitable variable involve in the implementation.
For completeness purpose, add the suitable parameters that are needed for each operation. Remember that the same operation of the different classes must have the same parameters.

(10 marks)